Class Meeting Handout #15

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We have seen a two-parameter plot in the Yukawa-Tsuno equation that separated inductive and resonance electronic effects of substituents. We will now explore a two-parameter system that separates electronic and steric effects as we consider the Taft equation. We will further define other common LFER schemes used for interpreting reaction mechanisms.[1]

The Taft Plot and Measuring "Size"

The Hammett plot is a cheat.[2] The steric demands are held constant because the changing substituents are on the far side of the phenyl ring, away from the reaction centre. Robert Taft introduced a two-parameter system[3] for more "real-world" systems where the substituent is not isolated from the reaction centre.[4,5]

Pre-Class Exercise: A Reality Check – 10 minutes

Not every reaction in organic chemistry includes a *meta* or *para*-substituted benzene ring. An example of another set of possibilities is *ortho*-substituted systems.

1. Compare the plots that you prepared using the data from the pre-class exercise with your peers. What did you observe?

Exercise: Finding the Taft Parameters – 30 minutes

Examining the method used by Taft and others to determine steric parameters based on reaction kinetics will help you to better understand the Taft equation.

2. Plot the data for acid hydrolysis of substituted ethyl benzoates[6] presented in table 1 on page 3 against the Hammett σ values. What is the reaction constant, ρ, for acid-catalyzed benzoate ester hydrolysis? Is there a significant substituent electronic effect in this process?

3. Taft used the reaction constant for base-catalyzed hydrolysis of substituted ethyl benzoates as the "standard ρ^* value" for all ester hydrolysis reactions. Plot rate data for this reaction and determine the reaction constant, ρ^*. We will use this value as our standard value. Table 2 collects some rate data from Ingold[7] and Evans.[8]

4. Write out the mechanism for acid and base-catalyzed hydrolysis of an ester. Use these mechanisms to explain the observed reaction constants, ρ, in each case above.

This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle*. Plots were typeset using *MatPlotLib* and *Python* tools.

[1] Read chapter 8.3 and 8.4.

[2] Sorry to burst your bubble. Also, "the cake is a lie!"

[3] See ch. 8.4.1

[4] "Polar and Steric Substituent Constants for Aliphatic and o-Benzoate Groups from Rates of Esterification and Hydrolysis of Esters." R.W. Taft, *J. Am. Chem. Soc.*, **1952**, *74*, 3120–3128. https://doi.org/10.1021/ja01132a049

[5] "Linear Free Energy Relationships from Rates of Esterification and Hydrolysis of Aliphatic and *ortho*-Substituted Benzoate Esters." R.W. Taft, *J. Am. Chem. Soc.*, **1952**, *74*, 2729–2732. http://doi.org/10.1021/ja01131a010

Figure 1: Substituted ethyl benzoate and ethyl acetate. Which will have a steric effect? ↓

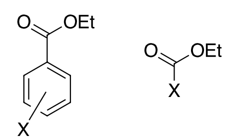

Figure 2: The Taft equation. ↓

$$log\frac{k_X}{k_{CH_3}} = \rho^*\sigma^* + \delta E_s$$

[6] "The activation energy of organic reactions. Part III. The kinetics of acid hydrolysis of esters." E.W. Timm, C.N. Hinshelwood, *J. Chem. Soc.*, **1938**, *168*, 862–869. http://doi.org/10.1039/jr9380000862

[7] "Mechanism of, and constitutional factors controlling, the hydrolysis of carboxylic esters. Part VIII. Energies associated with induced polar effects in the hydrolysis of substituted benzoic esters." K.C. Ingold, W.S. Nathan, *J. Chem. Soc.*, **1936**. 222–225. https://doi.org/10.1039/jr9360000222

[8] "Studies of the ortho-effect. Part III. Alkaline hydrolysis of benzoic esters." D.P. Evans, J.J. Gordon, H.B. Watson, *J. Chem. Soc.*, **1937**, 1430–1432. https://doi.org/10.1039/jr9370001430

5. Use the values for the acid hydrolysis of substituted ethyl acetates in table 4 or 5 to determine your own Taft E_s steric parameters[9] Compare them to the literature values presented in table 7.[10]

6. We have determined a ρ^* value for base-catalyzed benzyl ester hydrolysis and can now assign this value as Taft's standard ρ^* value for aliphatic ester hydrolysis. Use the rate constants for base hydrolysis in table 4 to complete the Taft equation for base hydrolysis of an aliphatic ester and thus determine a set of σ^* values for the substituent groups. How do they compare to the literature values?[11]

7. Taft and other established a list of substituent parameters, σ^* for groups that are directly attached to the reaction centre. Are these generally applicable? Plot the pK_a values for various carboxylic acids and alcohols vs. the Taft σ^* values and the Hammett σ values.[12] See tables 3, 6 and 7 for data.

[9] Set $E_s = 0$ for the CH_3 group and use and $\delta = 1$ for ester hydrolysis as the standard reaction, just like Taft did.

[10] from "Free Energy Relationships in Organic and Bio-organic Chemistry" by A. Williams, **2003**, pg 272. https://doi.org/10.1039/9781847550927.

[11] Sound impossible? Remember that, for aliphatic esters, you now have a set of E_s values from the acid hydrolysis data. The value of δ is still set at one because ester hydrolysis, base or acid, is the reference reaction for steric effects. We have now set the standard reaction constant, ρ^*, to the same value as for benzoate esters so that the Taft substituent parameters will have a relationship with the Hammett parameters. We are "scaling" the σ^* parameters to better match the Hammett σ parameters. We have the rates and can easily calculate the relative rates w.r.t. the methyl group, and so we will have values for $\log {}^{k_X}/_{k_{Me}}$. Now we have every part of the Taft equation, except for values of σ^*. We can now easily solve the Taft equation for σ^* in each case.

[12] Should you use σ_m, σ_p, σ^+ or σ^-? They were all determined using benzene rings as a sterically constant spacer.

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will...

- have learned about steric parameters and how they are defined in the Taft equation.
- be able to define the Taft σ^* parameters for electronic effects and describe how they compare to the corresponding Hammett σ values.
- have used experimental data to create your own Taft E_s parameters.
- **Computer Skill**: be able to use an interactive *Python* notebook to interpret and analyze experimental data from the literature.

A Look Ahead

We now have a scoring system for the electronic effect of a substituent and a scoring system for the steric effect of a substituent. By plotting an appropriate set of σ values against the log of the observed rate, we can determine the sensitivity of the transition state to changes in electron density and obtain a value for ρ. We can also use E_s steric parameters combined with σ^* electronic parameters to determine the sensitivity of the transition state for steric crowding and determine the δ value for the reaction centre.

At the next meeting, we will present scoring systems for solvent effects and for nucleophilicity. Measuring the sensitivity of a transition state to these parameters also can provide information to support a mechanistic hypothesis.

Drug discovery is driven in a large part by exploring "quantitative structure-activity relationships" (QSAR) in which these scoring systems and others are used to determine and predict effects of structural changes on drug efficacy. We can explore this idea further if you wish.[13]

[13] I hope you are making a "wish list" for topics to explore in the second half of this course. We can explore an idea of your choosing using the tools of physical organic chemistry. I am willing to replace any planned example with something in which you are actually interested.

Data Tables

The activities in this handout are supported by the data in the following tables. The data sets can be accessed directly using the given links to the GitHub repository.

Substituent	k /10^{-5} s^{-1}
p-OCH_3	6.17
p-OH	4.47
p-CH_3	7.76
H	9.12
p-Cl	7.94
p-Br	8.31
p-NO_2	10.7
m-NO_2	8.91
o-NO_2	0.513

Table 1: Rate of acid-catalyzed solvolysis for ethyl benzoates in 50% ethanol, 100 °C.
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This data is from reference 6 and can be obtained from my GitHub repository at https://raw.githubusercontent.com/blinkletter/4410PythonNotebooks/refs/heads/main/Class_17/data/17-Table_1.csv

Substituent[8]	k /10^{-5} $\frac{\text{mol}}{\text{L s}}$	Substituent[7]	k /10^{-5} $\frac{\text{mol}}{\text{L s}}$
p-NH_2	1.27	H	62.1
p-OCH_3	11.5	m-CH_3	43.3
p-CH_3	25.1	m-Cl	477
H	55.0	m-NO_2	4290
p-Cl	237	p-F	126
p-I	278	p-NO_2	7200
p-Br	289		
p-NO_2	5670		

Table 2: Rate of base-catalyzed solvolysis for ethyl benzoates in 85% ethanol, 25 °C.
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This data is from references 7 and 8 and can be obtained from my GitHub reporitory at https://raw.githubusercontent.com/blinkletter/4410PythonNotebooks/refs/heads/main/Class_17/data/17-Table_2.csv

Substituent (X)	pK_a (XCOOH)	pK_a (XCH_2OH)
CH_3	4.76	16
CH_2Cl	2.86	14.31
CH_2Br	2.86	
$CHCl_2$	1.29	12.89
CCl_3	0.65	12.24
CBr_3	0.66	
CF_3	0	12.37
CH_2F	2.66	
C_6H_5	4.31	15.4
CH_2CN	2.43	
CH_2OCH_3	3.53	
C_2F_5		11.35

Table 3: pK_a values for selected carboxylic acids and alcohols, pK_a values obtained from the Williams pK_a compilation [PDF]
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The data in this table can be obtained from my GitHub repository at https://raw.githubusercontent.com/blinkletter/4410PythonNotebooks/refs/heads/main/Class_17/data/17-Table_4.csv

Acyl Substituent (X)	Acid Hydrolysis k_{obs} / 10^{-6} s^{-1}			Base Hydrolysis k_{obs} / 10^{-2} $M^{-1}s^{-1}$		
H,	747	±	155	3230	±	893
CH_3,	43.0	±	5.9	11.2	±	2.0
CH_2Cl,	27.8	±	3.2			
C_2H_5,	36.6	±	1.7	5.49	±	0.63
n-C_3H_7,	18.8	±	0.9	2.57	±	0.53
n-C_4H_9,	17.5	±	2.0	2.18	±	0.45
n-C_5H_{11},	17.1	±	2.8	3.16	±	0.15
i-C_5H_{11},	19.2	±	0.4			
CH_2Ph,	17.9	±	1.7	16.6	±	1.1
CH_2CH_2Ph,	16.0	±	3.3	8.11	±	0.19
$CH_2CH_2CH_2Ph$,	15.3	±	2.1	4.46	±	0.21
i-C_3H_7,	14.6	±	2.0	1.27	±	0.24
c-C_6H_{11},	6.97	±	2.73			
i-C_4H_9,	5.05	±	1.16	0.645	±	0.089
CH_2−c-C_6H_{11},	4.50	±	0.52	0.833		
CHMeEt,	3.19	±	1.17			
CHMePh,	2.78	±	0.45	1.32		
CHPhEt,	1.36	±	0.31	0.426		
t-C_4H_9,	1.24	±	0.06	0.047	±	0.013
$CHPh_2$,	0.75	±	0.26	0.910		
$CHEt_2$,	0.45	±	0.29	0.022	±	0.003
CH=CHPh,	0.44			1.99	±	0.41
CCl_3,	0.38					
Ph,	0.12	±	0.22	0.976	±	0.090

Table 4: Taft's experimental values for rates of acid and base hydrolysis of aliphatic esters. Data is from reference 5.
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Note: Taft reported relative values. To convert them to rate constants, I used the reported values for acid-catalyzed hydrolysis of ethyl acetate from reference 6 and the value for base-catalyzed from "The Reaction Rate of the Alkaline Hydrolysis of Ethyl Acetate." H. Tsujikawa, H. Inoue, *Bull. Chem. Soc. Japan*, **1966**, *39*, 1837–1842, https://doi-org.proxy.library.upei.ca/10.1246/bcsj.39.1837

The data can be obtained at my GitHub repository at https://raw.githubusercontent.com/blinkletter/4410PythonNotebooks/refs/heads/main/Class_17/TaftData1952rates.csv.

The *Python* notebook that describes how I ~~faked~~ processed this data can be found at https://github.com/blinkletter/4410PythonNotebooks/blob/main/Class_17/taftData1952Processing.ipynb

Substituent	k /10^{-5} s^{-1}
CH_3	1072
CH_2Cl	724
$CHCl_2$	275
CCl_3	550

Table 5: Rate of acid-catalyzed solvolysis for ethyl acetates in 50% ethanol, 100 °C.[6]
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This data can be obtained from my GitHub repository at https://raw.githubusercontent.com/blinkletter/4410PythonNotebooks/refs/heads/main/Class_17/data/17-Table_3.csv

Substituent	σ_o	σ_m	σ_p	σ^+	σ^-
NH_2		−0.16	−0.66	−1.3	−0.15
Me	0.29	−0.06	−0.14	−0.31	−0.17
H	0	0	0	0	0
OMe		0.11	−0.27	−0.78	−0.26
OH		0.12	−0.37	−0.92	−0.37
F		0.34	0.06	−0.17	−0.03
I	1.34	0.35	0.18	0.14	0.27
Cl	1.26	0.37	0.23	0.11	0.19
Br	1.35	0.39	0.23	0.15	0.25
CN	1.06	0.56	0.66		0.88
NO_2	2.03	0.72	0.78		1.24

Table 6: Selected Hammett substituent constants that may be useful in interpreting the data in this handout.
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This data is from reference 10 and can be obtained from my GitHub repository at https://raw.githubusercontent.com/blinkletter/4410PythonNotebooks/refs/heads/main/Class_17/data/LFER_Williams.csv

Substituent (X)	E_s	σ^*
H	1.24	0.49
Me	0	0
Et	−0.07	−0.1
Pr	−0.36	−0.115
i-Pr	−0.47	−0.19
n-Bu	−0.39	−0.13
i-Bu	−0.93	−0.21
t-Bu	−1.54	−0.30
CEt_3	−3.8	
CH_2Cl	−0.24	1.05
$CHCl_2$	−1.54	1.94
CCl_3	−2.06	2.65
CH_2Br	−0.27	1.00
$CHBr_2$	−1.86	1.96
CBr_3	−2.43	2.43
CH_2F	−0.24	1.10
CF_3	−1.16	2.61
$CH_2C_6H_5$	−0.38	0.215
CH_2CN	−0.94	1.30
CH_2OCH_3		0.52
C_2F_5		2.56
C_6H_5	−2.48	0.60

Table 7: Selected Taft E_s steric and σ^* electronic parameters. Data is from reference 10.
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The full data set can be obtained from my GitHub repository at https://raw.githubusercontent.com/blinkletter/LFER-QSAR/refs/heads/main/data/Taft_Es_Williams.csv